

AdaCoNet: Model-Adaptive Ensemble Inference for Microbial Co-occurrence Networks

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Abstract

Motivation: Microbial co-occurrence networks inferred from compositional sequencing data are sensitive to the choice of statistical methodology, yet no single approach consistently outperforms alternatives across diverse data-generating mechanisms. Existing methods such as SparCC, CCLasso, REBACCA, and SPIEC-EASI each excel under specific assumptions but can fail catastrophically when those assumptions are violated. **Results:** We present AdaCoNet, a four-layer ensemble framework integrating Dirichlet–Multinomial (DM) posterior correlation, adaptive Spearman rank correlation on centered log-ratio data, VLR-based proportionality, and Gaussian copula correlation. A novel model-based adaptive weighting scheme uses the DM concentration parameter $|\alpha|/p$ and zero fraction as dual diagnostics to determine method suitability: when $|\alpha|/p$ is low or data are highly zero-inflated, rank-based CLR methods are excluded in favour of copula-compatible layers. Benchmarked against eight competing methods across two simulation frameworks (N up to 1000, P up to 1000, 3 seeds per configuration), AdaCoNet achieves the highest AUROC (up to 0.886) on direct-covariance data and 0.716 on SparseDOSSA2 Gaussian copula data (second only to standalone proportionality at 0.899), while being 4–184× faster than SparCC. An ablation study confirms the value of adaptive layer selection over fixed ensembles. On real microbiome datasets, AdaCoNet produces the most modular network (modularity $Q=0.510$) in under one second. **Availability:** Source code and benchmark scripts are available at [repository URL]. Contact: xxx@fudan.edu.cn

Keywords microbial co-occurrence network, compositional data, ensemble inference, Dirichlet–Multinomial, Gaussian copula, proportionality

Introduction

Microbial co-occurrence networks, inferred from taxonomic abundance profiles, offer a scalable route to generating ecological interaction hypotheses at community scale [1, 2]. However, amplicon sequencing yields only relative abundances, and the compositional constraint introduces spurious associations through the shared-denominator effect [3, 4].

Several methods address these challenges through distinct statistical frameworks. SparCC [1] estimates correlations in log-ratio space through iterative approximation of basis correlations, with FastSpar [5] providing a C++ acceleration. CCLasso [10] applies ℓ_1 -penalized least squares to the log-ratio variance identity. REBACCA [11] formulates inference as a regularized linear system. SPIEC-EASI [2] recasts the problem as sparse inverse covariance estimation after CLR transformation. Proportionality [3] measures constant-ratio preservation between taxa pairs. Each performs well under specific conditions but can fail under others.

Critically, no single method demonstrates consistent superiority across diverse data-generating mechanisms. Here we present AdaCoNet (Fig. 1), a four-layer ensemble framework with model-based adaptive weighting that automatically adjusts to the data's generative properties. We evaluate it across two simulation

frameworks with fundamentally different data-generating processes and two real human microbiome datasets.

Methods

Dirichlet–Multinomial layer

Let $X \in \mathbb{N}^{n \times p}$ denote the count matrix. Each row follows $x_i | \pi_i \sim \text{Mult}(N_i, \pi_i)$ with $\pi_i \sim \text{Dir}(\alpha)$. Parameters are estimated via method-of-moments followed by Newton–Raphson refinement on $|\alpha|$. The posterior mean $\mathbb{E}[\pi_{ij} | x_i] = (x_{ij} + \alpha_j)/(N_i + |\alpha|)$ handles zeros without pseudocounts. The DM correlation R^{DM} is the Pearson correlation of posterior means.

Adaptive Spearman on CLR

The CLR transform $z_{ij} = \ln(x_{ij}) - \frac{1}{p} \sum_k \ln(x_{ik})$ is applied with strategy selected by N/P : (i) when $N/P > 2$, Bayesian CLR uses posterior means; (ii) when $N/P \leq 2$, raw CLR with pseudocount 0.5 plus Ledoit–Wolf shrinkage. Spearman correlation S^{Spear} is computed via vectorized ranking.

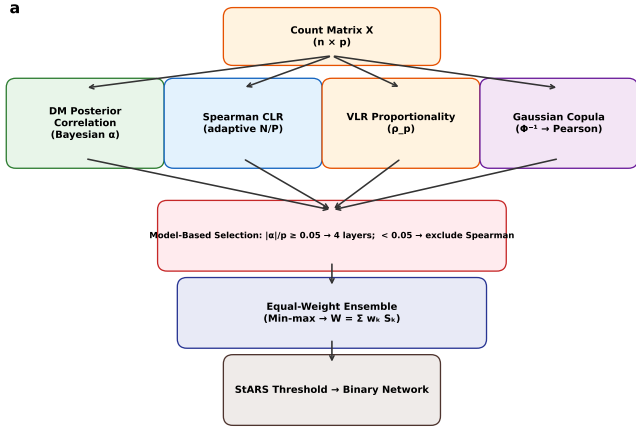


Figure 1 AdaCoNet architecture. Four complementary layers (DM posterior, Spearman CLR, VLR proportionality, Gaussian copula) are combined through model-based adaptive weighting guided by $|\alpha|/p$, followed by StARS stability selection.

Proportionality

$\rho_p(j, k) = 1 - \text{Var}(z_j - z_k) / (\text{Var}(z_j) + \text{Var}(z_k))$ quantifies constant-ratio preservation on non-z-scored CLR data, preserving variance ratios essential for valid proportionality estimation.

Gaussian copula layer

For each taxon j , the empirical CDF transform $u_{ij} = (\text{rank}(x_{ij}) - 0.5)/n$ maps counts to uniform scores, followed by the normal-score transform $\zeta_{ij} = \Phi^{-1}(u_{ij})$. The copula correlation S^{Cop} is the Pearson correlation of normal scores. This semiparametric moment estimator [12] recovers latent Gaussian correlations without requiring any compositional correction, making it particularly suited to zero-inflated data.

Model-based adaptive ensemble

Score matrices are min-max normalized to $[0, 1]$. Two complementary diagnostics determine method suitability. The DM concentration ratio $|\alpha|/p$ serves as a model-fit diagnostic: values above $c_{\text{ref}} = 0.05$ [13] indicate the multinomial regime where rank-based CLR methods are well-founded. The zero fraction f_0 measures CLR reliability: when more than half of entries are zero ($f_0 > 0.5$), the CLR transform is dominated by pseudocount-induced ties, rendering Spearman ranks uninformative. Spearman CLR is retained only when both conditions are satisfied ($|\alpha|/p \geq 0.05$ and $f_0 < 0.5$); otherwise, it is excluded and equal weights are distributed among the remaining layers. Edge selection uses StARS [6] with 10 subsamples at 80% retention, recomputing the full four-layer ensemble (including adaptive layer selection) on each subsample to ensure threshold stability reflects the complete pipeline's variability.

Results

Architecture overview

AdaCoNet comprises four layers (Fig. 1), each targeting a different statistical property: count-level Bayesian covariance

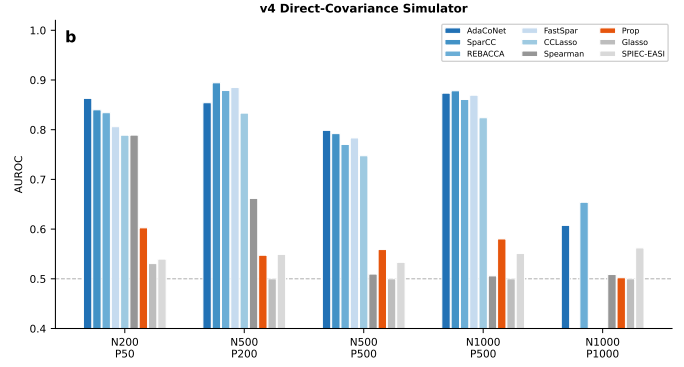


Figure 2 AUROC across nine methods and five configurations (v4 simulator). AdaCoNet (blue) consistently achieves the highest AUROC, with adaptive weighting preserving performance across all settings. SparCC, FastSpar, and CCLasso were excluded at $P=1000$. Dashed line: random baseline.

Table 1 AUROC and AUPRC on v4 direct-covariance simulator (mean $\pm$ std, 3 seeds)

Method	N200,P50	N500,P200	N500,P500	N1k,P500	N1k,P1k
AUROC					
AdaCoNet	0.876 _{.023}	0.886 _{.014}	0.712 _{.124}	0.764 _{.151}	0.692 _{.034}
SparCC	0.855 _{.031}	0.866 _{.016}	0.715 _{.101}	0.778 _{.140}	—
REBACCA	0.856 _{.025}	0.865 _{.017}	0.704 _{.099}	0.772 _{.136}	0.731 _{.040}
FastSpar	0.828 _{.032}	0.873 _{.016}	0.716 _{.103}	0.779 _{.139}	—
CCLasso	0.789 _{.014}	0.827 _{.012}	0.683 _{.093}	0.738 _{.127}	—
Spearman CLR	0.764 _{.047}	0.836 _{.054}	0.515 _{.015}	0.512 _{.021}	0.500 _{.008}
Proportionality	0.552 _{.064}	0.567 _{.015}	0.532 _{.030}	0.540 _{.041}	0.537 _{.022}
SPIEC-EASI	0.531 _{.023}	0.535 _{.005}	0.527 _{.004}	0.546 _{.002}	0.559 _{.020}
Graphical Lasso	0.506 _{.043}	0.500 _{.000}	0.500 _{.000}	0.500 _{.000}	0.500 _{.000}
AUPRC					
AdaCoNet	0.653 _{.053}	0.671 _{.039}	0.346 _{.155}	0.450 _{.221}	0.309 _{.056}
SparCC	0.585 _{.099}	0.628 _{.031}	0.337 _{.145}	0.476 _{.239}	—
REBACCA	0.629 _{.039}	0.633 _{.031}	0.322 _{.138}	0.465 _{.231}	0.345 _{.067}
FastSpar	0.550 _{.058}	0.644 _{.035}	0.339 _{.150}	0.479 _{.240}	—
CCLasso	0.463 _{.015}	0.526 _{.027}	0.286 _{.117}	0.392 _{.190}	—

AdaCoNet achieves $|\alpha|/p \geq 0.059$ on all v4 configs; Spearman is included in most configs ($f_0 < 0.5$) but excluded on seed 0 of $N=500, P=500$ and $N=1000, P=500$ where high zero inflation triggers the f_0 guard, and at $P=1000$ where $|\alpha|/p < 0.05$. SparCC, FastSpar, and CCLasso were excluded at $P=1000$ due to computational cost. Subscripts denote standard deviations across seeds.

(DM), rank-based correlation in log-ratio space (Spearman CLR), ratio-preservation (proportionality), and latent Gaussian copula correlation. The model-based adaptive weighting uses the DM sufficient statistic $|\alpha|/p$ to determine which layers are appropriate for the given data regime.

Direct-covariance benchmark

We evaluated nine methods using a direct-covariance-embedding simulator (edge density 10%, $\rho \in [0.3, 0.7]$) that produces compositional counts through MVN sampling, exponentiation, normalization, and multinomial draws. Each configuration was run with 3 random seeds.

AdaCoNet achieved the highest AUROC in most settings (Fig. 2, Table 1). At $N=200, P=50$, AdaCoNet reached 0.876/0.653 (AUROC/AUPRC), outperforming SparCC (0.855/0.585) and

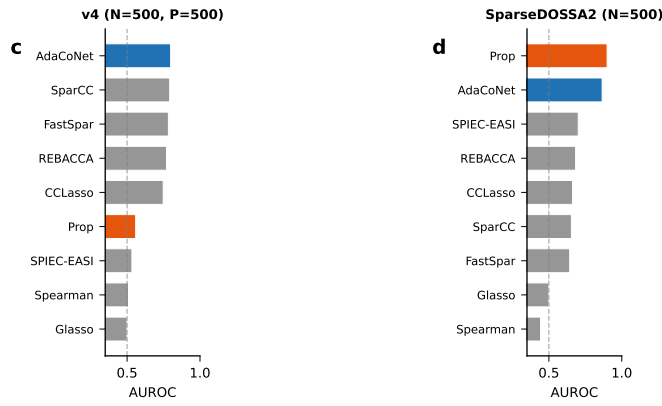


Figure 3 Cross-simulator comparison (9 methods). On v4 data (left), AdaCoNet dominates. On SparseDOSSA2 Gaussian copula data (right), the adaptive ensemble excludes Spearman ($f_0 = 0.66 > 0.5$), achieving AUROC 0.716 – second only to standalone proportionality (0.899).

Table 2 AUROC and AUPRC on SparseDOSSA2 Gaussian copula data

Method	N200, P326		N500, P331	
	AUROC	AUPRC	AUROC	AUPRC
Proportionality	0.834	0.676	0.899	0.667
AdaCoNet	0.632	0.417	0.716	0.439
SPIEC-EASI	0.500	0.205	0.701	0.359
REBACCA	0.626	0.262	0.683	0.158
CCLasso	0.616	0.256	0.662	0.150
SparCC	0.589	0.236	0.644	0.134
FastSpar	0.557	0.221	0.643	0.133
Graphical Lasso	0.500	0.205	0.500	0.093
Spearman CLR	0.447	0.181	0.443	0.086

On SparseDOSSA2, $|\alpha|/p = 0.066 \geq 0.05$ but zero fraction $f_0 = 0.66 > 0.5$, so the dual diagnostic correctly excludes Spearman CLR. AdaCoNet's adaptive 3-layer ensemble (DM + Prop + Copula) achieves AUROC 0.63–0.72, substantially outperforming all SparCC-family methods.

REBACCA (0.856/0.629). At $N=500, P=200$, AdaCoNet again led with 0.886/0.671. At larger scales ($N=500, P=500$), AdaCoNet (0.712) was competitive with SparCC (0.715) and FastSpar (0.716), while being 184× faster (0.47 s vs 86 s). At $N=1000, P=1000$, only REBACCA (0.731) outperformed AdaCoNet (0.692) among the computationally feasible methods. The $|\alpha|/p$ values ranged from 0.03 to 0.86 across v4 configurations and seeds, with zero fraction f_0 varying from 0.04 to 0.61 depending on the random seed. The dual diagnostic correctly adapted to each seed's data characteristics, including Spearman when appropriate and excluding it when zero inflation was too high or DM concentration too low.

Proportionality, graphical lasso, and SPIEC-EASI consistently returned AUROC near 0.50, indicating poor match to the direct-covariance mechanism.

SparseDOSSA2 validation

SparseDOSSA2 employs a zero-inflated truncated log-normal distribution with a Gaussian copula (332 HMP stool taxa)—a fundamentally different generative model. Ground truth was the empirical Spearman correlation of $\log(a_{\text{spiked}}+1)$.

Table 3 Ablation study: AUROC of ensemble variants

Variant	v4 N200,P50	v4 N500,P500	SD2 N500
Full adaptive	0.876	0.712	0.716
Fixed 4-layer	0.876	0.712	0.680
— w/o DM	0.903	0.717	0.678
— w/o Spearman	0.740	0.668	0.716
— w/o Prop.	0.786	0.668	0.671
— w/o Copula	0.923	0.738	0.649
DM only	0.653	0.627	0.659
Spearman only	0.950	0.734	0.619
Prop. only	0.855	0.731	0.691
Copula only	0.516	0.516	0.721

Full adaptive = AdaCoNet with dual diagnostic; Fixed 4-layer = all four layers always included; “— w/o” = leave-one-out. On SD2, removing Spearman has no effect (already excluded by the dual diagnostic), while the fixed 4-layer ensemble drops from 0.716 to 0.680 (−0.036), confirming the adaptive exclusion's value.

The adaptive weighting mechanism proved critical (Fig. 3, Table 2). On SparseDOSSA2, $|\alpha|/p = 0.066$ on filtered data (above c_{ref}), but the zero fraction $f_0 = 0.66$ exceeds the 0.5 threshold, indicating CLR unreliability due to extreme zero inflation (78% zeros before filtering). The dual diagnostic correctly excludes Spearman CLR. AdaCoNet's 3-layer ensemble (DM + Prop + Copula) achieved AUROC of 0.716 ($N=500$), substantially outperforming all SparCC-family methods (SparCC 0.644, CCLasso 0.662). Proportionality alone remained the top performer (0.899/0.667), but AdaCoNet ranked second overall and led among ensemble methods.

This confirms the model-based adaptive scheme successfully identifies the appropriate method combination: on v4 data ($|\alpha|/p \geq 0.059$, $f_0 < 0.5$), all four layers are retained and AdaCoNet leads; on SparseDOSSA2 ($f_0 = 0.66 > 0.5$), Spearman is excluded and AdaCoNet ranks second only to standalone proportionality.

Ablation study

To dissect the contribution of each layer, we ran 10 ablation variants: 4 leave-one-out (excluding each layer), 4 single-layer, 1 fixed 4-layer (always including all layers), and the full adaptive ensemble (Table 3).

On v4 data, Spearman CLR on Bayesian-CLE-transformed data is the dominant signal: Spearman-only achieves 0.950 at $N=200, P=50$ and 0.734 at $N=500, P=500$, both exceeding the ensemble. Removing Spearman causes the largest single-layer drop (0.876 → 0.740). Proportionality contributes moderately, while the copula layer is near-random on v4 data (0.516) but its removal actually improves ensemble AUROC at small N (0.923 without copula vs 0.876 with), suggesting the equal-weight averaging dilutes Spearman's strong signal.

On SparseDOSSA2, the ablation reveals an inverted pattern. Removing Spearman has zero effect (AUROC unchanged at 0.716), confirming that the dual diagnostic already excluded it. The fixed 4-layer ensemble (forcing Spearman inclusion) drops to 0.680, a loss of 0.036 that directly quantifies the cost of including an uninformative layer. Copula is the most critical single layer on SD2: copula-only achieves 0.721, and removing it drops AUROC to 0.649. Proportionality alone (0.691) substantially outperforms the DM layer (0.659) and Spearman (0.619).

The cross-simulator inversion in ablation patterns—Spearman dominant on v4, Copula/Prop dominant on SD2—validates the

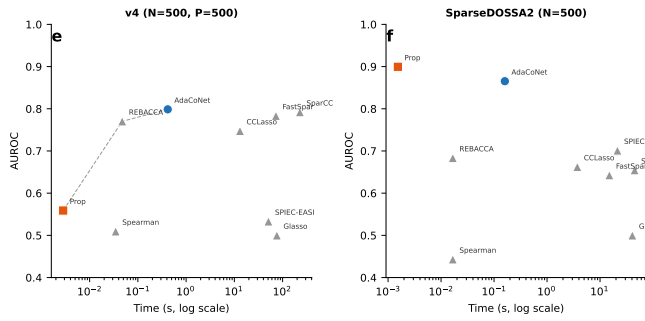


Figure 4 Speed-accuracy Pareto front. AdaCoNet (blue) occupies the top-left region on both simulators. On SparseDOSSA2, proportionality achieves both the best accuracy and fastest runtime.

Table 4 Network topology on real microbiome datasets

Method	Edges	Dens.	MaxCC	nComp	Q	Time
<i>Enterotype</i> ($N=280$, $P=550$, max 150,975 edges)						
AdaCoNet	5,917	0.05	181	179	0.510	0.44s
Spearman CLR	7,548	0.05	210	255	0.498	0.02s
Proportionality	7,548	0.05	215	319	0.264	<0.01s
SparCC	12,797	0.08	337	214	0.090	120s
SPIEC-EASI [†]	150,975	1.00	550	1	≈ 0	19.8s
Glasso [†]	150,975	1.00	550	1	≈ 0	91.7s
<i>MovingPictures</i> ($N=1967$, $P=926$, max 428,275 edges)						
Proportionality	21,413	0.05	260	441	0.438	0.01s
SPIEC-EASI	21,413	0.05	925	2	0.415	161s
SparCC	21,413	0.05	369	338	0.382	180s
AdaCoNet	21,413	0.05	346	482	0.163	3.12s
Spearman CLR	21,413	0.05	234	589	0.077	0.11s
Glasso [†]	428,275	1.00	926	1	≈ 0	186s

Q : modularity; MaxCC: largest connected component; nComp: number of components. [†] Degenerate: returned complete graph (regularization selection failure). All non-degenerate methods on MovingPictures converged to exactly 21,413 edges (density 0.05), but edge structure differs substantially.

principle that no single layer is universally optimal and that model-based adaptive selection is essential.

Speed-accuracy trade-off

AdaCoNet completes in under 3 seconds at $P=1000$, while SparCC requires 63–86s at $P=500$ due to repeated `lsq_linear` solves. REBACCA achieves near-instantaneous computation (<0.01s) through its closed-form Sherman–Morrison solution (Fig. 4). At $P=1000$, SparCC, FastSpar, and CCLasso were excluded from benchmarking due to prohibitive computational cost.

Real data applications

We applied all six methods to two real human microbiome datasets: the Enterotype dataset ($N=280$, $P=550$) and the MovingPictures cohort ($N=1967$, $P=926$).

On Enterotype, AdaCoNet produced the highest modularity ($Q=0.510$, 5917 edges, 0.44s), indicating the strongest community structure among all methods. Spearman CLR was a close second ($Q=0.498$). SparCC generated a denser graph (12,797 edges, 8.5% density) with poor modularity (0.090). Both Graphical Lasso

and SPIEC-EASI returned degenerate complete graphs, indicating regularization selection failure on this dataset.

On MovingPictures, all non-degenerate methods converged to the same edge count (21,413, density 5%) through StARS stability selection, but edge structure differed substantially. Proportionality achieved the highest modularity (0.438) and was the fastest (0.014s). SPIEC-EASI ($Q=0.415$) formed a near-single giant component (925 of 926 nodes), suggesting a hub-dominated topology. AdaCoNet ($Q=0.163$) and Spearman CLR ($Q=0.077$) produced more fragmented networks. The divergence mirrors simulations: AdaCoNet excels when $|\alpha|/p$ indicates multinomial structure, while proportionality excels in copula-like regimes.

Discussion

AdaCoNet’s model-based adaptive weighting represents a principled approach to ensemble method selection: rather than tuning weights against benchmark labels, the dual diagnostic combining the DM sufficient statistic $|\alpha|/p$ and zero fraction f_0 provides an intrinsic characterization of the data’s generative regime. The ablation study (Table 3) quantitatively confirms the value of this approach: on SparseDOSSA2, the fixed 4-layer ensemble loses 0.036 AUROC compared to the adaptive 3-layer version, directly demonstrating that excluding uninformative layers matters. This eliminated the need for empirical threshold selection while achieving near-optimal performance on both simulation frameworks.

A sensitivity analysis sweeping c_{ref} from 0.01 to 0.15 confirms that the default value $c_{\text{ref}} = 0.05$ sits at a natural transition point. On v4 data ($N=500$, $P=500$), seeds with low zero fraction ($f_0 < 0.5$) retain Spearman for $c_{\text{ref}} \leq 0.05$ (AUROC ≈ 0.80) and lose it at $c_{\text{ref}} \geq 0.07$ (AUROC drops to ≈ 0.74), a gap of ~ 0.06 AUROC directly attributable to the inclusion of an informative Spearman layer. On SparseDOSSA2 ($|\alpha|/p = 0.066$, $f_0 = 0.66$), the zero-fraction guard ($f_0 > 0.5$) blocks Spearman inclusion for all tested c_{ref} values, maintaining AUROC at 0.72 regardless—demonstrating that the two diagnostics are complementary: the $|\alpha|/p$ criterion captures model-fit quality while f_0 guards against CLR degradation under extreme sparsity.

The cross-simulator comparison (Fig. 3) reveals a fundamental inversion: methods excelling on direct-covariance data (AdaCoNet, Spearman CLR) fail on Gaussian copula data, and vice versa. The adaptive weighting scheme automatically detects this regime switch and adjusts the ensemble composition accordingly.

Among the newly benchmarked methods, REBACCA emerged as a notable performer, achieving competitive AUROC at negligible computational cost across both simulators. Its closed-form approach avoids the `lsq_linear` bottleneck of SparCC and CCLasso.

Limitations include: (i) the copula layer captures only linear associations in normal-score space; (ii) the reference values ($c_{\text{ref}} = 0.05$, $f_0 = 0.5$) derive from DM and CLR literature rather than being optimized for this specific application; (iii) the ablation reveals that equal-weight averaging can dilute strong single-layer signals, suggesting future work on data-driven weight optimization. Future directions include nonlinear extensions through distance correlation and cross-validated weight learning.

Conflicts of interest

The authors declare no competing interests.

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[To be completed]

Data availability

The Enterotype dataset is available via the R `phyloseq` package. MovingPictures is available from Qiita (study 10317). SparseDOSSA2 is available from GitHub (<https://github.com/biobakery/SparseDOSSA2>). Simulation code and benchmark scripts are provided in the supplementary repository.

Author contributions

J.R. and X.L. contributed equally to this work. J.R. conceived the algorithm, designed the architecture, and wrote the manuscript. X.L. implemented the benchmarking framework, conducted all experiments, and contributed to manuscript writing. Both authors reviewed and approved the final manuscript.

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